60 single-best-answer questions | 90 minutes | 60 marks

Answer all questions. Choose one answer per question. Use a separate answer sheet.

1. A severe rise in intracranial pressure produces systemic hypertension with reflex bradycardia. What is this response?

A. Lung hypoxia reflex.
B. Atrial stretch reflex.
C. Bainbridge reflex.
D. Cushing reflex.

2. Which pressure relationship helps prevent fluid accumulation in normal pulmonary alveoli?

A. Pulmonary capillary pressure consistently exceeds systemic arterial pressure.
B. Absence of plasma proteins in pulmonary capillaries.
C. Relatively low pulmonary capillary hydrostatic pressure.
D. Higher alveolar protein pressure than in capillary plasma.

3. A patient in compensated haemorrhagic shock has cold, pale skin. Which mechanism accounts for this?

A. Increased sympathetic cutaneous vasoconstriction.
B. Loss of vasomotor drive causing widespread dilation.
C. Increased skin capillary blood volume from vasodilation.
D. Direct activation of warm-sensitive skin receptors.

4. A cuff that is far too small is used on a large upper arm. Which measurement error is likely?

A. No systematic effect on cuff pressure.
B. An undersized cuff gives a falsely high reading.
C. Only diastolic pressure becomes unmeasurable.
D. Falsely low BP from excessive arterial unloading.

5. The CSF volume is about 150 mL and production is 550 mL/day. Approximately how many complete turnovers occur per day?

A. About 4.
B. About 2.
C. About 1.
D. About 6.

6. Which recording satisfies the basic chapter criteria for a normal sinus rhythm at rest?

A. Rate 72/min, equal R-R gaps, identical preceding P waves, PR 0.16 s.
B. Rate 72/min, equal R-R gaps, no visible P waves, PR not measurable.
C. Rate 72/min, unequal R-R gaps, identical preceding P waves, PR 0.16 s.
D. Rate 72/min, equal R-R gaps, identical preceding P waves, PR 0.26 s.

7. Increasing capillary hydrostatic pressure while other Starling forces stay unchanged tends to produce what?

A. Decreased outward fluid filtration.
B. More fluid filtration into tissues.
C. Reversal of lymphatic flow.
D. Increased plasma oncotic attraction.

8. A patient develops cardiac tamponade. Why can stroke volume fall despite a normal blood volume?

A. Tamponade increases functional ventricular compliance.
B. Tamponade directly raises venous return above normal.
C. The aortic valve cannot open due solely to a high heart rate.
D. High pericardial pressure restricts ventricular filling.

9. Why are superficial neck veins often less distended in a healthy person who stands upright?

A. Venous pressure decreases above the heart.
B. The venae cavae reverse their direction of blood flow.
C. Systemic venous pressure exceeds aortic systolic pressure.
D. The skull's arteries pull venous valves fully open.

10. Arterial pressure inside an arteriole is unchanged, but pressure in its surrounding tissue rises. What happens to transmural pressure?

A. It decreases because the pressure difference narrows.
B. It increases because the pressure difference widens.
C. It becomes identical to intravascular arterial pressure.
D. It remains fixed because arterial pressure is unchanged.

11. A severely underperfused patient begins breathing rapidly, although the lungs were initially healthy. What drives this change?

A. Increased aerobic ATP production stimulates breathing.
B. Suppression of the peripheral chemoreceptors stimulates breathing.
C. Lactic acidosis stimulates peripheral chemoreceptors.
D. High tissue oxygen content increases respiratory drive.

12. Why may advanced liver disease cause generalised oedema even without venous obstruction?

A. Excess albumin synthesis draws water into tissues.
B. Less albumin synthesis reduces plasma oncotic pressure.
C. A rise in plasma protein concentration stops absorption.
D. Liver failure directly shuts every lymphatic valve.

13. How does local hypoxia affect most systemic arterioles versus pulmonary arterioles?

A. Systemic vessels constrict while pulmonary vessels dilate.
B. Both primarily constrict to lower oxygen delivery.
C. Both remain relatively insensitive to tissue oxygen.
D. Systemic vessels dilate; pulmonary vessels constrict.

14. A pressure tracing shows atrial contraction shortly after an ECG P wave and ventricular contraction shortly after QRS. What principle explains both delays?

A. Electrical excitation normally precedes the mechanical response of cardiac muscle.
B. Mechanical contraction normally finishes before electrical activation begins in the chamber.
C. The T wave must occur before either chamber can begin contraction.
D. Atrial and ventricular pressures directly create the corresponding ECG waves.

15. A student faints after receiving distressing news, with a slow pulse and low arterial pressure. Which mechanism best matches the lecture?

A. Severe medullary ischaemia causing a Cushing pressor response.
B. Emotional inhibition of pressor activity with vagal predominance.
C. Metabolic stimulation of the heart's sympathetic beta-1 receptors.
D. Atrial volume-receptor firing after excessive fluid infusion.

16. What extra perspective do the chest leads V1-V6 add to the standard limb-lead views?

A. They replace voltage measurement with intracardiac pressure.
B. They remove the need to record the frontal electrical plane.
C. They isolate atrial electrical activity from ventricular activity.
D. They show cardiac electrical vectors in the horizontal plane.

17. During progressive shock, capillary hydrostatic pressure paradoxically rises while systemic arterial pressure is falling. Which microvascular pattern best explains it?

A. Precapillary constriction with freely draining venules.
B. Increased venular drainage with reduced capillary entry.
C. Open capillary entry with constricted venules.
D. Precapillary constriction with widened venous outlets.

18. A stiff ventricle has high filling pressure but a preserved ejection fraction. What is the primary defect?

A. Impaired relaxation and filling.
B. Increased contractility and ejection.
C. Reduced electrical activation of the SA node.
D. Excessive ventricular compliance.

19. Lead II has an upright QRS while aVR has a downward QRS during the same beat. What best explains the difference?

A. Each lead projects the same electrical vector onto a different axis.
B. One lead measures depolarisation and the other repolarisation.
C. The two leads record different mechanical contractions.
D. Atrial and ventricular activity alternate between the leads on each beat.

20. A patient with marked polycythaemia has elevated haematocrit. What is the expected direct effect on systemic flow resistance?

A. Lower viscosity and lower total vascular resistance.
B. Higher viscosity and lower resistance.
C. Higher viscosity and higher resistance.
D. Lower viscosity and higher resistance.

21. Why can a brief complete interruption of cerebral blood flow cause unconsciousness within seconds?

A. The brain has very limited stores of oxygen and fuel.
B. The brain stores enough ATP for several minutes of activity.
C. The cerebral circulation can switch immediately to venous oxygen.
D. The brainstem normally stops all electrical activity first.

22. A resting muscle suddenly becomes active while systemic arterial pressure stays steady. What most directly increases its blood supply?

A. Its large veins reduce their compliance.
B. Its arteries increase their driving pressure.
C. Its local arterioles reduce their resistance.
D. Its local capillaries develop stronger contractions.

23. At one instant, Lead I records +0.4 mV and Lead III records -0.2 mV. What must Lead II record?

A. -0.2 mV, from using the third lead alone.
B. +0.6 mV, from adding their magnitudes.
C. +0.2 mV, from adding the signed voltages.
D. -0.6 mV, from subtracting the two values.

24. Compared with a large artery, why can a large vein store more blood for a small rise in pressure?

A. Venous walls contain more elastic recoil tissue than aorta.
B. Veins have high compliance and distensible walls.
C. Venous vessels are made mostly of cardiac muscle.
D. Venous pressure generally exceeds aortic systolic pressure.

25. A region of heart muscle starts working harder. Which local product of its own metabolism helps increase its blood flow?

A. Serotonin released by nearby platelets.
B. Angiotensin II formed in the blood.
C. Noradrenaline from sympathetic nerves.
D. Adenosine formed as ATP is used up.

26. What physiological purpose is served by slow conduction through the AV node?

A. It prevents the impulse from ever reaching the ventricular myocardium.
B. It ensures ventricles eject before the atria contract.
C. It allows the aortic valve to remain open during atrial systole.
D. It allows atrial contraction to finish ventricular filling.

27. Why is a patient with significant coronary narrowing more vulnerable to angina when heart rate rises sharply?

A. The P wave disappears and blocks blood flow.
B. Myocardial oxygen consumption falls below baseline.
C. The coronary sinus develops arterial hypertension.
D. Shorter diastole and higher myocardial demand.

28. Which statement best reflects the updated interpretation of capillary fluid balance?

A. The glycocalyx regulates filtration; lymph returns excess fluid.
B. Lymphatic vessels play no role when hydrostatic pressure is normal.
C. Most capillaries continuously reabsorb nearly all filtered fluid at their venous ends.
D. Interstitial proteins do not affect exchange at any capillary.

29. Minutes after a severe allergic exposure, a patient's pressure drops with generalised swelling. Which vascular action is most relevant?

A. Histamine-related dilatation with increased capillary permeability.
B. Generalised constriction with reduced capillary permeability.
C. Increased myocardial pump function with reduced venous capacitance.
D. Selective haemorrhage into the gastrointestinal tract.

30. After standing suddenly, a healthy person has a transient fall in end-diastolic volume. Which cause is most direct?

A. Increased atrial contraction generating excess preload.
B. Increased circulating red-cell mass.
C. Greater coronary artery blood flow.
D. Venous pooling reducing return to the heart.

31. How does normal inspiration tend to assist systemic venous return?

A. It increases right atrial pressure and reduces abdominal pressure.
B. It closes all systemic venous valves simultaneously.
C. It increases peripheral venous capacitance until return stops.
D. Lower chest pressure draws venous blood toward the heart.

32. An exercising leg receives increased blood flow while its muscle continues contracting. What is this response called?

A. Active hyperaemia driven by ongoing tissue metabolism.
B. Reactive hyperaemia following a temporary flow block.
C. Baroreflex vasodilation after systemic arterial pressure rises.
D. Autoregulation that holds its flow at rest level.

33. Two cylindrical vessels have equal transmural pressure, but vessel X has twice the radius of vessel Y. How do their required wall tensions compare?

A. Both vessels require exactly the same wall tension.
B. Vessel X requires half the wall tension of Y.
C. Vessel X requires four times the tension of Y.
D. Vessel X requires twice the wall tension of Y.

34. Why does persistent lactic acidosis make circulatory shock harder to reverse?

A. It prevents further fluid filtration from capillaries.
B. It depresses contraction and catecholamine responses.
C. It directly restores aerobic ATP production.
D. It increases ventricular force and vascular responsiveness.

35. Which of these locally released substances dilates arterioles?

A. Histamine
B. Endothelin-1
C. Serotonin
D. Angiotensin II

36. A previously anxious patient with severe prolonged shock becomes apathetic and poorly responsive. Which explanation is most concerning?

A. Red-cell production has rapidly restored oxygen delivery.
B. Skin vasoconstriction has resolved the shock.
C. The skeletal-muscle pump has become overactive.
D. Worsening cerebral ischaemia impairs brain function.

37. An ECG machine runs at 25 mm/s. How much time passes while the paper advances by 1 mm horizontally?

A. 0.10 s.
B. 0.50 s.
C. 0.20 s.
D. 0.04 s.

38. A cerebral vessel ruptures and blood enters brain tissue. Which type of cerebrovascular event is present?

A. Cerebral hypoglycaemia.
B. Embolic ischaemia.
C. Thrombotic ischaemia.
D. Haemorrhagic stroke.

39. A person lies down after standing upright. What happens to pulmonary blood volume in the lecture's illustration?

A. It falls by approximately 400 mL.
B. It remains fixed regardless of posture.
C. It rises by approximately 400 mL.
D. It falls until alveolar flow stops.

40. A ventricle has an end-diastolic volume of 130 mL and an end-systolic volume of 55 mL. What is its stroke volume?

A. 130 mL.
B. 75 mL.
C. 55 mL.
D. 185 mL.

41. How does CSF substantially reduce the effective weight of the brain?

A. It increases the brain's intrinsic density.
B. It provides buoyant support against gravity.
C. It reduces blood volume to near zero.
D. It directly reduces the brain's oxygen consumption.

42. In a whole-body Fick measurement, which blood should the return-side oxygen sample represent?

A. Oxygenated blood in a systemic artery.
B. Mixed venous blood entering the lungs.
C. Blood in a single peripheral venule.
D. Capillary blood from active skeletal muscle only.

43. Why is 1500 used to calculate rate from the number of small squares between R waves at 25 mm/s?

A. It is the number of 1-mm squares passing in one minute.
B. It is the number of millimetres in one normal cardiac cycle.
C. It is the electrical voltage calibration in millivolts.
D. It is the standard number of R waves recorded in one minute.

44. A patient with massive haemorrhage remains severely hypotensive after blood volume is completely restored. Which stage fits?

A. Early compensated hypovolaemic shock with intact reflexes.
B. Irreversible shock after extensive tissue injury.
C. Postural hypotension from remaining upright.
D. Transient vasovagal faint without major tissue damage.

45. After a period of low blood volume, aldosterone secretion increases. Which effect is most important over the following hours to days?

A. Greater renal sodium loss with accompanying water loss.
B. Immediate electrical acceleration of the intrinsic cardiac pacemaker.
C. Greater renal sodium retention with accompanying water retention.
D. Immediate widening of resistance vessels across the circulation.

46. A bleeding patient has a nearly normal arterial pressure but cold skin and reduced urine output. What is the best conclusion?

A. Normal pressure excludes significant shock.
B. Low perfusion can coexist with normal pressure.
C. Cardiac output must be above its pre-bleeding level.
D. Blood flow to all organs must remain normal.

47. Why can chronic widespread alveolar hypoxia lead to pulmonary hypertension?

A. Persistent hypoxia opens every pulmonary capillary without resistance.
B. Persistent hypoxic constriction increases lung resistance.
C. Widespread hypoxia lowers pulmonary vascular smooth-muscle tone.
D. Chronic hypoxia eliminates blood passing through the lungs.

48. A sphygmomanometer cuff is held substantially below heart level. How is its pressure reading affected by gravity?

A. It reads falsely low because of reduced hydrostatic pressure.
B. It reads falsely high because of added hydrostatic pressure.
C. It is unaffected because arterial vessels have valves.
D. It measures venous instead of arterial pressure.

49. Once local metabolic dilation restores an organ's blood flow, what prevents dilation from continuing unchecked?

A. The restored flow removes oxygen from the tissue cells.
B. The restored flow causes a sustained rise in local lactate.
C. The restored flow clears accumulated dilator metabolites.
D. The restored flow abolishes vessel-wall smooth muscle.

50. A graph of organ flow against perfusion pressure is almost flat over a middle range but slopes downward at very low pressures. Why?

A. The arterioles reach their maximal dilation at low pressure.
B. The arterioles reach their maximal constriction at low pressure.
C. The heart stops driving any pressure into the organ.
D. The veins become the principal resistance at low pressure.

51. A ventricular impulse occurs independently of atrial excitation after complete AV conduction block. What has happened?

A. The SA node has increased its firing rate.
B. The aortic valves transmit an alternate impulse.
C. Atrial excitation now travels directly through the lungs.
D. A subsidiary escape pacemaker has taken over.

52. A burn patient presents with shock and an initially high haematocrit. What explains the contrast with recent whole-blood haemorrhage?

A. Burns selectively destroy red cells circulating in blood.
B. Burns produce increased circulating volume despite shock.
C. Burns cause immediate erythropoietin-driven erythrocytosis.
D. Plasma leakage leaves intravascular red cells concentrated.

53. The jugular venous 'c' wave is chiefly linked to which event?

A. Rapid emptying of the right atrium.
B. Passive systemic venous filling late in systole.
C. Systolic bulging of the tricuspid valve.
D. Right atrial contraction immediately before QRS.

54. After three minutes upright, pressure changes from 126/82 to 104/77 mmHg. How is this classified using the chapter criterion?

A. No orthostatic hypotension because diastolic pressure fell only 5 mmHg.
B. No orthostatic hypotension because final systolic pressure exceeds 100 mmHg.
C. Orthostatic hypotension because systolic pressure fell 22 mmHg.
D. Orthostatic hypotension only if both pressures fall by 20 mmHg.

55. When both atrioventricular valves are open and semilunar valves are shut, which process predominates?

A. Rapid ventricular ejection into the aorta.
B. Ventricular filling during diastole.
C. Closure of the outflow valves at end systole.
D. Isovolumetric ventricular contraction.

56. Following regional node removal, a patient develops persistent firm limb swelling. What explains it?

A. Raised arterial oxygen causes tightness.
B. Systolic ejection fraction necessarily becomes reduced.
C. Protein accumulation after failed lymph drainage.
D. Excess venous flow clears local proteins.

57. Which feature most reliably distinguishes first-degree AV block from second-degree AV block in the chapter's basic ECG classification?

A. Both have identical PR intervals on each beat.
B. Both show independent P-wave and QRS rhythms.
C. First-degree conducts every P; second-degree drops some.
D. First-degree lacks P waves; second-degree lacks QRS complexes.

58. A blood-pressure reading is 132/84 mmHg. What is the pulse pressure?

A. 36 mmHg.
B. 44 mmHg.
C. 48 mmHg.
D. 40 mmHg.

59. Under the Frank-Starling law, why does greater ventricular filling strengthen systolic force?

A. Reduced preload stimulates immediate calcium extrusion.
B. Stretch of myocardial fibres strengthens contraction.
C. Lower venous return increases sarcomere overlap.
D. Higher afterload increases systemic venous return.

60. A corrected QT interval is 0.46 s. Relative to the upper limit shown in the ECG diagram, how should it be described?

A. Below the displayed upper limit of 0.40 s.
B. Exactly equal to the displayed upper limit of 0.46 s.
C. Above the displayed upper limit of 0.44 s.
D. Within the displayed upper limit of 0.50 s.